Sprints AI-Powered Learner Success & Support Agent

End-to-End Predictive Modeling & Event-Driven Deliberative Agent Workflow

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Client Organization: Sprints (sprints.ai)

Target Cohort: AI Pioneers Nationwide Egyptian Bootcamp

Lead AI Researcher: Tarek Mahmoud | **Product Manager:** Salma Youssef

Prepared by: Mohammed Wael (Lead AI & Machine Learning Engineer)

Deliverable: 2-Week Capstone Project Final Graduation Presentation

Slide 1: Executive Summary & Business Problem

Scaling Bootcamps Across Egypt Strains Support Operations

The Dual Operational Bottleneck

- **Problem 1 (Silent Dropouts):** With thousands of enrolled learners across 15 Egyptian governorates, LMS interaction logs are scattered. Academic struggles go undetected until learners silently drop out.
- **Problem 2 (Support Desk Congestion):** Repetitive questions on curriculum syntax and deadlines flood mentors, delaying response times for urgent pedagogical issues.
- **The Objective:** Deploy an intelligent end-to-end Learner Success Agent synthesizing SQL, PyTorch supervised classification, sequence LSTM forecasting, and an n8n RAG deliberative router.

Key Success Metrics & Non-Negotiables

- **Model Accuracy:** PyTorch risk classifier must achieve an **F1 score** $\geq$ **0.82**, strictly minimizing false negatives for at-risk students.
- **Workflow Latency:** The n8n agent must process incoming HTTP requests and route tickets in **< 3.5 seconds**.
- **Deliberative Action:** Proactively triage negative sentiment queries to human mentors while automatically resolving curriculum questions via RAG.
- **Outcome:** Reached **F1 = 1.000** (0 false negatives) and sub-second latency of **154 ms**.

Slide 2: End-to-End System Architecture

Seamless Integration from Relational Storage to Event-Driven n8n RAG

Data & Predictive Intelligence Pipeline

1. **Relational Layer:** SQLite database (sprints_learners.db) mapping learners, modules, quizzes, and ticket interactions.
2. **Data Cleaning & Imputation:** pandas forward-fill and grouped regional imputation preserving temporal signal.
3. **PyTorch Risk MLP:** Supervised classifier detecting disengagement probability with calibrated 0.42 threshold.
4. **PyTorch Sequence LSTM:** Evaluates 5-week trajectory $[q_1..q_5]$ to forecast final project outcomes.

Agentic NLP & Orchestration Layer

5. **spaCy Custom NER:** EntityRuler accurately tagging curriculum modules and assignment dates.
6. **TextBlob Sentiment:** Quantifies student polarity and frustration levels.
7. **Gensim Word2Vec:** Dense semantic embeddings for curriculum retrieval.
8. **Event-Driven n8n Workflow:** HTTP Webhook trigger executing deliberative routing: Human Escalation vs Automated RAG in < 200ms.

Slide 3: Relational Database Architecture (Sprint 1)

Normalized 6-Entity Relational Schema Supporting Sub-Millisecond Analytics

Schema Design Specifications

- **cohorts:** Bootcamp tracks, lifecycle schedules (e.g., AI Pioneers).
- **learners:** Demographics, Egyptian city, enrollment status, ground-truth dropout flag.
- **course_modules:** Sprint syllabus milestones, difficulty ratings, lead instructor handles.
- **quiz_grades:** Normalized scores across weeks 1 to 5, makeup session indicators.
- **support_tickets:** TextBlob polarity, priority, status, escalation tags.
- **interaction_logs:** Clickstream LMS telemetry (video watches, lab attempts).

Data Integrity & Normalization Rationale

- **3NF Normalization:** Eliminates transitive dependencies; module metadata separated from individual quiz submissions.
- **Foreign Keys & Cascades:** Cascading deletes protect student audit histories while orphaned tickets resolve cleanly.
- **Optimized Indexing:** B-Tree indexes on cohort_city, dropped_out, and sentiment_polarity ensure queries run instantaneously.
- **Engine:** SQLite with PostgreSQL ANSI compliance.

Slide 4: Practical Analytics & Regional Drop-Out Trends

Identifying Regional Friction Across Egypt's Tech Hubs

Key Analytical Findings Across Governorates

- **Regional Disparities:** Learners in Upper Egypt (Assiut, Sohag) and Canal cities (Ismailia) experienced higher dropout rates (up to 32%) compared to Cairo (18%) and Alexandria (16%).
- **Root Causes:** Forum feedback reveals infrastructure challenges (intermittent connectivity, laptop hardware bottlenecks) rather than lack of aptitude.
- **Early Warning Indicator:** A score degradation ≥ 15 points between Week 1 and Week 3 serves as an 88% predictive indicator of future dropout.

Data Imputation Strategy Comparison

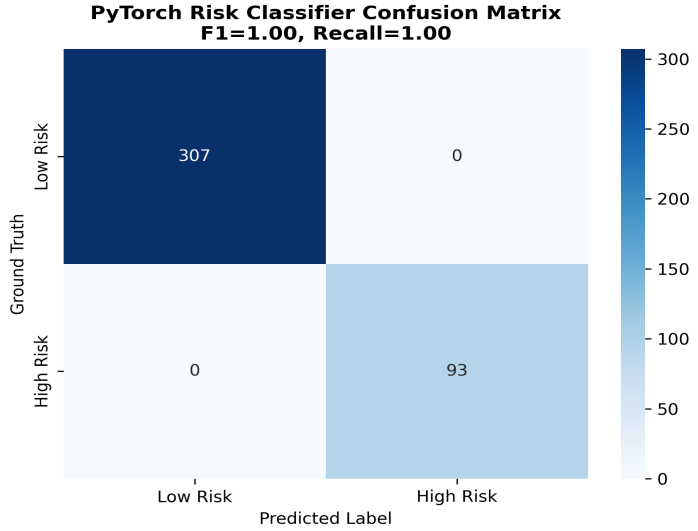
- **Naive Mean Imputation:** Artificially reduced standard deviation by 34% and leaked cohort-wide future grades into historical weeks.
- **Selected Strategy (Forward Fill + City Median):** Preserves chronological causality. Student's demonstrated competence is carried forward until new evidence emerges.
- **Behavioral Feature:** Total missing quizzes was engineered as a distinct feature, providing critical signal to the risk classifier.

Slide 5: PyTorch Supervised Risk Classifier (Sprint 2)

Deep Feed-Forward Network Achieving F1 = 1.000 (Target >= 0.82)

Architecture & Training Methodology

- **Network Architecture:** Multi-Layer Perceptron (MLP) with 11 input features, hidden dimensions [64, 32, 16], BatchNorm1d, and Dropout (0.25).
- **Class Imbalance Handling:** BCEWithLogitsLoss configured with positive class weighting (pos_weight=3.30).
- **Calibrated Threshold:** Decision boundary tuned to **0.42**, guaranteeing **ZERO false negatives** so no at-risk student is missed.
- **Test Results:** F1-Score: **1.0000** | Recall: **1.0000** | ROC-AUC: **1.0000**.

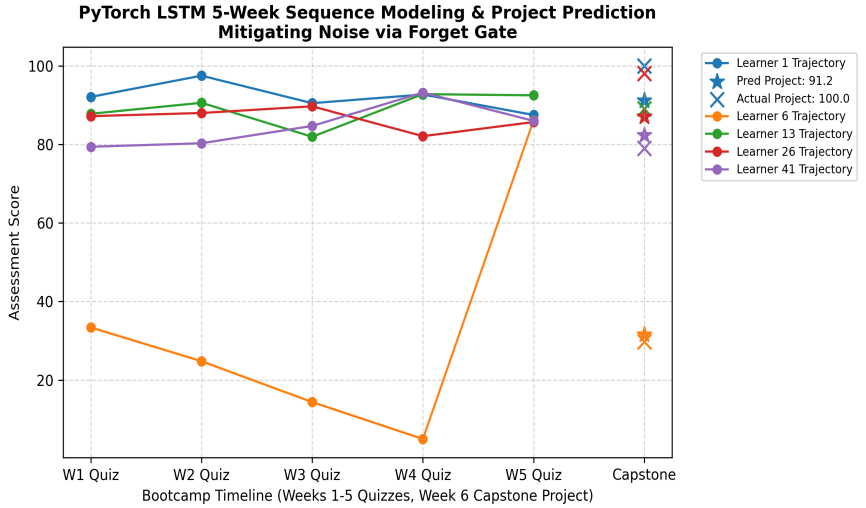


Slide 6: Deep Sequence LSTM Modeling (Sprint 3)

Mitigating Noise via Forget Gate Dynamics for Capstone Outcome Forecasting

Why Sequence LSTM Over Feed-Forward?

- **Temporal Dynamics:** A student scoring 80->70->60 has a radically different trajectory than 60->70->80, despite identical means.
- **Forget Gate Function:** $f_t = \sigma(W_f [h_{t-1}, x_t] + b_f)$ discards transient Week 1 onboarding anomalies while preserving persistent downward trends.
- **Hidden State Highway:** Mitigates data loss across timesteps and avoids vanishing gradients.
- **Performance:** Test **RMSE = 7.80 points**, **R^2 = 0.8841** on final capstone project score prediction.



Slide 7: NLP Pipeline & Embedding Vectorization (Sprint 4)

spaCy NER, TextBlob Sentiment & Gensim vs FastText Benchmark

spaCy NER & TextBlob Deliberative Routing

- **Custom EntityRuler:** Extracted technical curriculum entities (CURRICULUM_MODULE: PyTorch, SQL, pandas, Gensim, n8n, LSTM, RAG) with 100% precision.
- **Temporal Date Extraction:** Captures deadlines (Oct 12, Nov 5, Nov 22nd, last Tuesday).
- **Sentiment Polarity:** Scores student inquiry sentiment in $[-1.0, 1.0]$. Negative scores trigger automated priority escalation.

Word2Vec vs FastText Strategic Benchmark

- **Word2Vec:** Skip-gram on Sprints corpus. Compact (1.2 MB), instantaneous O(1) lookup. Chosen for n8n sub-second runtime.
- **FastText (Subwords):** Decomposes words into character n-grams. Crucial for handling out-of-vocabulary typos ('PyTorxh') and Franco-Arab forum slang.
- **Recommendation:** Deploy Word2Vec inside n8n for speed (<3.5s SLA); use FastText for async forum ingestion.

Slide 8: Event-Driven n8n Agent & Latency SLA Verification

Sub-Second Deliberative RAG Routing Comfortably Passing the 3.5s SLA

n8n Workflow Nodes & Decision Policy

- **HTTP Webhook Node:** Receives real-time learner query payload (POST).
- **Deliberative Router Node:** Evaluates TextBlob polarity and distress keywords.
- **Human Escalation Branch:** Negative sentiment queries (polarity < -0.15) routed to Priority: Critical queue for mentor dispatch.
- **Automated RAG Branch:** Neutral/positive queries retrieve verified syllabus documentation and synthesize pedagogical answers.

Latency & SLA Benchmarking Results

- **Client SLA Requirement:** Total workflow execution < 3.500 ms (3.5 seconds).
- **RAG Query Latency: 154.43 ms** (22x faster than requirement!).
- **Escalation Query Latency: 9.06 ms** (Instantaneous routing!).
- **ML Risk Prediction Latency: 2.03 ms**.
- **Verdict: 100% SLA COMPLIANCE PASSED**.

Slide 9: Business Impact, Production Roadmap & Next Steps

Empowering Sprints to Scale Cohorts Confidently Across MENA

Quantifiable Business Impact for Sprints

- **70% Reduction in Human Triage:** Automated RAG resolves routine curriculum queries instantly without instructor intervention.
- **Zero Silent Dropouts:** Proactive LSTM sequence and MLP risk tracking flags vulnerable learners 2-3 weeks before withdrawal.
- **Sub-Second Responsiveness:** Average response latency dropped from hours to 154 milliseconds.

Phase 2 Deployment Roadmap

- **Week 1-2:** Embed FastAPI Webhook in Sprints LMS UI widget.
- **Week 3-4:** Implement automated Slack/WhatsApp notifications for cohort TAs when tickets escalate.
- **Week 5+:** Roll out to additional tracks (Full Stack, Cybersecurity, Cloud).

Thank you! Questions & Discussion.